5.2: Write test cases and traceability

5.2.1. Requirements Traceability Matrix (RTM)

The Traceability Matrix ensures that every Functional Requirement (FR) documented in the SRS is covered by corresponding Use Cases (UC) and executable Test Cases (TC). This guarantees comprehensive test coverage and prevents scope creep.

Req. ID	Functional Requirement Description	Use Case Mapping	Test Case ID	Test Priority
FR-01	Account registration with email validation	UC-01	TC-001	High
FR-02	User login and authentication handling	UC-02	TC-002	High
FR-04	Display course catalog (code, name, schedule)	UC-03	TC-003	Medium
FR-05	Allow student enrollment in available courses	UC-04	TC-004	High
FR-06	Reject overlapping class schedules (Conflict Check)	UC-12, UC-04, UC-09	TC-005	Critical
FR-08	Generate and group weekly student timetable	UC-05	TC-006	High

FR-09	Track assignments/exams with dynamic status labels	UC-06	TC-007	High
FR-11	Allow enrolled students to post questions	UC-07	TC-008	Medium
FR-13	Allow lecturer to add/delete class sessions	UC-09	TC-009	High
FR-15	Allow lecturer to view, filter, and answer questions	UC-11	TC-010	High

5.2.2. Core Test Case Specifications

Below are the detailed test cases derived directly from the Acceptance Criteria (AC). These are written in a standard BDD (Behavior-Driven Development) Given/When/Then structure optimized for both manual execution and future test automation.

Test Case ID: TC-001 **Related Req:** FR-01 (AC-01) **Test Scenario:** Validate user registration behavior with new and existing email addresses.

- **Pre-conditions:** The visitor is on the Registration page and has no active session.
- **Test Steps:**
  - Enter a valid name, an unregistered email, and a valid password.
  - Select role as "Student" and click 'Register'.
  - Navigate back to the Registration page.
  - Attempt to register again using the exact same email address.
- **Expected Results:**
  - On step 2, the system successfully creates the account, establishes a session, and redirects to the dashboard.
  - On step 4, the system rejects the submission, displays a clear "Email already registered" error message, and does not create a duplicate account.

Test Case ID: TC-005 **Related Req:** FR-06 (AC-08) **Test Scenario:** Verify the schedule conflict detection algorithm during student enrollment.

- **Pre-conditions:** The student is logged in. The student is already enrolled in Course A with a Monday session from 07:00 to 09:00.
- **Test Steps:**
    - Navigate to the Course Catalog.
    - Select Course B, which has a scheduled session on Monday from 08:30 to 10:00, and click 'Enroll'.
    - Select Course C, which has a scheduled session on Monday from 09:00 to 11:00 (back-to-back), and click 'Enroll'.
- **Expected Results:**
    - For Course B (Step 2), the system blocks the enrollment, displaying a conflict error message detailing the overlap with Course A.
    - For Course C (Step 3), the system successfully processes the enrollment because back-to-back sessions are explicitly allowed by business rule BR-01.

Test Case ID: TC-007 **Related Req:** FR-09 (AC-11) **Test Scenario:** Verify chronological sorting and status labeling of assignments.

- **Pre-conditions:** The student is logged in and enrolled in a course containing assignments due yesterday, due today, due in 2 days, and due next week.
- **Test Steps:**
    - Navigate to the "Assignments" tab.
    - Observe the labeling and the sorting order of the unfinished items.
- **Expected Results:**
    - Items are dynamically labeled exactly as: "Overdue", "Due today", "Due soon" (for the item due in 2 days), and "Upcoming" (for next week).
    - The list is sorted in ascending order of the due date, ensuring the most urgent unfinished tasks appear at the top of the view.

Test Case ID: TC-010 **Related Req:** FR-15 (AC-14) **Test Scenario:** Verify lecturer authority in answering Q&A threads and status transition.

- **Pre-conditions:** The lecturer is logged in and assigned to a course. The course has existing student questions (some answered, some unanswered).
- **Test Steps:**
    - Navigate to the specific Course Q&A forum.
    - Identify an unanswered question and submit a reply text.
    - Check the status of the question after submission.
- **Expected Results:**
    - Unanswered questions are visually distinct from answered ones.
    - Upon saving the reply, the reply is visible to the class, and the question's status transitions permanently from "Unanswered" to "Answered".

5.2.1 Test approach and conventions

Scope

In scope: FR-01 to FR-16, BR-01 to BR-09 and NFR-01 to NFR-10. BR-05 duplicates BR-04, and BR-07 cannot be tested because the system has no edit or delete function for questions. Out of scope: F06-F09 (reminders, dashboard, attachments, LaTeX).

Test levels and methods

Level	Tool / environment	What it checks
Unit	Python unittest (test_app.py, AlgorithmTests)	Overlap check, timetable building, deadline classification and ranking
API	unittest + Flask test client with the in-memory store (EDUMATE_STORE=memory)	Permissions (401/403), business rules, error codes; no Firebase needed
UI / system	Manual, Chrome, Firebase test project	Screens, messages, end-to-end flows

Method codes in the tables: **A** = already automated in test_app.py; **A+** = easy to automate with the same pattern (a few lines); **M** = manual.

Priority, type and defect severity

- Priority: **Critical** = core business rule, a wrong result breaks the product; **High** = Must-have path; **Medium** = Should-have or secondary path; **Low** = quality checks that need extra effort.
- Type: P positive, N negative, B boundary, S security/permission, U usability, Perf performance.
- Severity: S1 blocks a main flow or corrupts data; S2 wrong result but a workaround exists; S3 cosmetic or wording.
- A test passes when the actual result equals the expected result; otherwise log a defect with its severity and TC ID.

Test accounts and data

Item	Value
Accounts (password Test@123)	gv1@test.vn (Lecturer GV1), gv2@test.vn (Lecturer GV2), sv1@test.vn (Student SV1), sv2@test.vn (Student SV2)
Courses	CS101 Mon 07:00-09:00 (GV1); MA101 Mon 08:30-10:00 (GV2); PH101 Mon 09:00-11:00 (GV1); EN101 Tue 07:00-09:00 (GV2); DB101 Wed 13:00-15:00 (GV1)
Overdue and boundary deadlines	Run load_sample.py after setting lecturer_email to gv1@test.vn in sample_data.json. Course CS102 (Tue 13:00-15:30, Thu 07:00-09:30) then has 'Bài tập 1' due 2 days ago and 'Bài tập 2' due in 3 days at 23:59. BR-06 blocks past dates in the UI, so an overdue item can only come from this script.

Environment, entry and exit criteria

- Environment: latest Chrome (also Edge and Firefox for NFR-09), local server (python app.py), Firebase test project with Email/Password enabled and Firestore created.
- Entry: the build runs, the accounts and courses above exist, and python -m unittest -v passes.
- Exit: all Critical and High cases pass, no open S1 defect, and every failed Medium or Low case has a logged defect.
- Smoke set (run first after every build, about 20 minutes): TC-001, 005, 010, 019, 020, 022, 024, 029, 033, 037, 041.

5.2.2 Requirements Traceability Matrix - functional requirements

Req. ID	Requirement (short)	Use case	Acceptance criteria	Test cases	Priority
FR-01	Register with name, email, password, role; reject duplicate email	UC-01	AC-01	TC-001, 002, 003, 004	High

FR-02	Log in; reject invalid credentials	UC-02	AC-02	TC-005, 006	High
FR-03	Log out; protected data blocked afterwards	UC-02	AC-03	TC-007, 008	High
FR-04	Show all courses with code, name, lecturer(s), schedule	UC-03	AC-04	TC-009	Medium
FR-05	Student enrolls in a course	UC-04	AC-05	TC-010, 011	High
FR-06	Reject overlapping sessions (enroll, add session, take over)	UC-04, UC-09, UC-13 (UC-12 is included)	AC-08, AC-10	TC-017, 019, 020, 021, 025	Critical
FR-07	Cancel enrollment; remove sessions and assignments from views	UC-04	AC-06	TC-012	High
FR-08	Weekly timetable grouped by weekday, sorted by start time	UC-05	AC-09	TC-022, 023	High
FR-09	Assignments and exams with status labels, unfinished first	UC-06	AC-11	TC-029, 030, 031	High
FR-10	Student marks an assignment as done	UC-06 (to be listed in PA2)	none yet (see 5.2.6)	TC-032	Medium
FR-11	Enrolled student posts a question	UC-07	AC-13	TC-037, 038, 039	High
FR-12	Lecturer creates a course	UC-08	AC-07	TC-013, 014, 015	High

FR-13	Lecturer adds and deletes class sessions	UC-09	AC-10	TC-024, 025, 026, 027, 028	High
FR-14	Lecturer adds and deletes assignments and exams	UC-10	AC-12	TC-033, 034, 035	High
FR-15	Lecturer views all questions and answers them	UC-11	AC-14	TC-040, 041, 042	High
FR-16	Lecturer takes over or steps down from a course	UC-13	none yet (see 5.2.6)	TC-016, 017, 018	Medium

Every FR is covered by at least one test case. System-level flows are covered by TC-054 and TC-055.

5.2.3 Requirements Traceability Matrix - business rules and non-functional requirements

Rule	Summary	Test cases
BR-01	Overlapping sessions rejected, back-to-back allowed	TC-019, 020, 021
BR-02	Assignments and Q&A only for enrolled students	TC-036, 038, 044
BR-03	Only the lecturer in charge manages sessions, deadlines and answers	TC-016, 028, 042
BR-04 / BR-05	Login required for all academic data	TC-007, 008
BR-06	Due date must be in the future	TC-035 (see 5.2.5)

BR-07	No edit or delete of an answered question	Not testable: the function does not exist; re-check if it is added
BR-08	Cancelling enrollment removes entries from the student's views	TC-012
BR-09	Answered only by a lecturer's reply	TC-041, 043 (see 5.2.5)
NFR-01, NFR-02	API response and page load time	TC-051
NFR-03	Token signature, expiry, audience, issuer checked	TC-045
NFR-04	401 without session, 403 for the wrong role	TC-008, 015, 046
NFR-05	No secrets in the repository	TC-047
NFR-06	First-time student finishes the main task in 3 minutes	TC-050
NFR-07	Vietnamese interface, one-sentence errors	TC-048
NFR-08	Failed operation leaves data unchanged	TC-052
NFR-09	Browser and screen-size compatibility	TC-049
NFR-10	Unit test coverage of the algorithms at least 80%	TC-053

5.2.4 Test case specifications

Columns: Req / AC = requirement and acceptance criterion covered; Type / Pri = test type and priority; Method = A, A+ or M (see 5.2.1). Messages in single quotes are the Vietnamese texts shown by the system.

F01 - Account Management

TC 1	Req /	Scenari	Preconditions data	Steps	Expected re	Type /	Method
TC-001	FR-01 / AC-01	Register a n student	Visitor signed c sv1@test.vn no registered	1) Open Register 2) E name 'SV One', the e password Test@123, r Student 3) Submit	Account creat user is signed and the main screen (Course tab) opens wit the user's nam	P / High	M
TC-002	FR-01 / AC-01	Register wit email that already exis	sv1@test.vn alr registered	1) Log out 2) Open Register 3) Enter the same email and subn	Error 'Email đ được đăng ký' second accou user stays on Register	N / High	M
TC-003	FR-01	Register as lecturer	gv1@test.vn no registered	Same as TC-001 with Lecturer	Account creat Courses tab sh the create-cou form that stud do not see	P / High	M

TC-004	FR-01	Invalid registration input	Visitor signed o	Try separately: (a) em name (b) email 'abc' (password '123'	Each is rejecte with a one-sentence Vietnamese message (e.g. 'Email không h lệ', 'Mật khẩu yếu (tối thiểu tự)'); no accou created	N / Medi	M
TC-005	FR-02 / AC-02	Log in with credentials	sv1@test.vn registered, sign out	Enter email and password, submit	Signed in; nam and role show tabs Courses, Timetable, Assignments, available	P / High	A (test_login_logou M
TC-006	FR-02 / AC-02	Log in with wrong credentials	sv1@test.vn registered	(a) correct email, wro password (b) unknow email	Error 'Sai ema hoặc mật khẩ both cases; no session; still o login screen	N / High	M
TC-007	FR-03 / AC-03, BR-04	Log out bloc protected d	SV1 signed in	1) Click Log out 2) Op /api/courses directly the browser 3) Press	Login screen shown; /api/courses returns 401 'C đăng nhập'; n course data v	S / High	A + M
TC-008	BR-04, NFR-04	Protected A without sess	No session	Call GET /api/course /api/timetable, /api/deadlines, /api/questions and P /api/courses/x/enro	Every call retu 401	S / High	A+

F02 - Course Registration

TC ID	Req / AC	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-009	FR-04 / AC-04	Course list content	CS101 and MA101 exist with sessions	1) Sign in as SV1 2) Open Courses 3) Repeat as GV1	Each card shows code, name, lecturer name(s), number of students and every session (weekday, time range, room), for both roles	P / Medium	M
TC-010	FR-05 / AC-05	Enroll in a course	SV1 not enrolled in CS101	Click Enroll on CS101	Success; card shows the enrolled state with a cancel option; CS101 sessions appear in Timetable	P / High	A (test_flow)
TC-011	FR-05	Enroll twice in the same course	SV1 already enrolled in CS101	Send the enroll request again (second tab or API)	Rejected 409 'Bạn đã đăng ký môn này'; still one enrollment	N / Medium	A (test_flow)

TC-012	FR-07 / AC-06, BR-08	Cancel enrollment	SV1 enrolled in CS101 (with sessions, one deadline marked done) and in PH101	1) Click Cancel enrollment on CS101 2) Open Timetable 3) Open Assignments	CS101 shown as not enrolled; its sessions gone from Timetable; its deadlines gone from Assignments; PH101 data unchanged	P / High	A (test_flow, partly) + M
TC-013	FR-12 / AC-07	Lecturer creates a course	GV1 signed in	Enter code 'cs101' and name 'Nhập môn lập trình', click Create	Course appears as CS101 (code in upper case) with GV1 as lecturer; sessions are added afterwards (TC-024)	P / High	A (test_flow)
TC-014	FR-12	Duplicate course code	CS101 exists	Create another course with code CS101	Rejected 409 'Mã môn học đã tồn tại'; no duplicate	N / Medium	A+
TC-015	FR-12, NFR-04	Student cannot create a course	SV1 signed in	Send POST /api/courses as SV1 (the UI hides the form)	403 'Bạn không có quyền thực hiện thao tác này'; no course created	S / High	A (test_flow)

TC-016	FR-16, BR-03	Lecturer takes over a course	DB101 created by GV1; GV2 has no session on Wed 13:00-15:00	1) GV2 clicks Take over on DB101 2) GV2 opens Q&A and the DB101 management area	Success; lecturer line shows both names; GV2 can see DB101 questions and add sessions and deadlines	P / Medium	A (second-lecturer test)
TC-017	FR-16, FR-06	Take over blocked by a schedule conflict	GV2 teaches MA101 (Mon 08:30-10:00); CS101 is Mon 07:00-09:00	GV2 clicks Take over on CS101	Rejected 409 'Trùng lịch dạy với MA101 (Thứ 2 08:30-10:00)'; GV2 not added	N / Medium	A (second-lecturer test)
TC-018	FR-16	Step down from a course	GV2 is co-lecturer of DB101; GV1 is the creator	(a) GV2 clicks Step down (b) GV1 tries to step down (API)	(a) Success, GV2 loses access (403 on DB101 sessions and questions) (b) Rejected 403 'Bạn là người tạo môn nên không thể thôi phụ trách'	P+N / Medium	A (second-lecturer test)

F03 - Automated Scheduling

TC ID	Req / AC	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-019	FR-06 / AC-08, BR-01	Overlapping session rejected on enrollment	SV1 enrolled in CS101 (Mon 07:00-09:00); MA101 is Mon 08:30-10:00	Click Enroll on MA101	Rejected 409 'Trùng lịch với CS101 (Thứ 2 07:00-09:00)'; the message names the clashing course and time; MA101 not enrolled	N / Critical	A (test_flow)
TC-020	FR-06 / AC-08, BR-01	Back-to-back sessions allowed (boundary)	SV1 enrolled in CS101; PH101 is Mon 09:00-11:00	Click Enroll on PH101	Success (end 09:00 equals start 09:00, not an overlap); both courses appear in Timetable	B / Critical	A (test_flow)
TC-021	FR-06	Same time on a different weekday allowed	SV1 enrolled in CS101 (Mon 07:00-09:00); EN101 is Tue 07:00-09:00	Click Enroll on EN101	Success; no conflict	B / High	A+

TC-022	FR-08 / AC-09	Timetable grouped and sorted	SV1 enrolled in CS101 and PH101 (Mon) and EN101 (Tue)	Open Timetable	Seven day columns Monday to Sunday; Monday lists CS101 before PH101; Tuesday lists EN101; other days empty; each slot shows time, course code and name, room	P / High	A (test_flow) + M
TC-023	FR-08	Timetable without enrollment	New student with no enrollment	Open Timetable	Seven empty days shown, no error	B / Low	M
TC-024	FR-13 / AC-10	Lecturer adds a session	GV1 in charge of CS101	Choose Monday, 07:00, 09:00, room A101, click Add	Session appears on the course card; enrolled students see it in their Timetable	P / High	A (test_flow)
TC-025	FR-13, FR-06 / AC-10	Session overlapping the lecturer's own schedule	GV1 teaches CS101 (Mon 07:00-09:00) and PH101	Add Mon 08:00-10:00 to PH101	Rejected 409 'Trùng lịch dạy với CS101 (Thứ 2 07:00-09:00)'; schedule unchanged	N / Critical	A (test_flow)

TC-026	FR-13	Invalid session time	GV1 in charge of CS101	Add a session with end equal to start, then one with end earlier than start	Rejected 400 'Giờ kết thúc phải sau giờ bắt đầu'; nothing saved	N / Medium	A+
TC-027	FR-13 / AC-10	Lecturer deletes a session	CS101 has a session; SV1 enrolled	GV1 clicks the delete icon on the session	Session removed from the course card and from SV1's Timetable	P / High	M
TC-028	FR-13, BR-03	Lecturer not in charge cannot change sessions	GV2 is not in charge of CS101	GV2 sends add and delete session requests for CS101 (API)	403 'Bạn không phụ trách môn học này'; no change; the UI shows no session controls on that course	S / High	A (second-lecturer test)

F04 - Deadline and Task Tracking

TC ID	Req / AC	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-029	FR-09 / AC-11	Status labels	SV1 enrolled in CS101 (items by GV1: due today 23:59, due in 2 days, due in 10 days) and in CS102	Open Assignments	Labels are Overdue, Due today, Due soon and Upcoming respectively (Vietnamese	P / High	A (classify test) + M

			('Bài tập 1' due 2 days ago)		equivalents in the UI)		
TC-030	FR-09 / AC-11	Sorting	Same data; mark the 'due today' item as done	Open Assignments	Unfinished items first in ascending due date (overdue at the top); the done item is listed after them	P / High	A (rank test) + M
TC-031	FR-09 / AC-11	Boundary: due in exactly 3 days	CS102 'Bài tập 2' due in 3 days at 23:59	Open Assignments	Per AC-11 the label is Due soon. The current code may show Upcoming (72-hour window), see 5.2.5	B / Medium	A+
TC-032	FR-10	Mark as done and undo	SV1 has an unfinished assignment	1) Tick the checkbox 2) Reload the page 3) Untick it	Item shows as done and moves below the unfinished ones; still done after reload; unticking restores it	P / Medium	A (test_flow, partly) + M

TC-033	FR-14 / AC-12	Lecturer adds an assignment and an exam	GV1 in charge of CS101; SV1 enrolled	Add 'BT1' (Assignment) due in 2 days 23:59 and 'Giữa kỳ' (Exam) due in 10 days	Both listed on the course with type and due date; SV1 sees both in Assignments	P / High	A (test_flow)
TC-034	FR-14 / AC-12	Lecturer deletes an item	An item exists and SV1 marked it done	GV1 deletes the item	Item disappears from the lecturer's list and from SV1's Assignments	P / Medium	M
TC-035	FR-14, BR-06	Past due date on creation	GV1 in charge of CS101	Add an item whose due date and time are in the past	Per BR-06 the item is rejected with a message and not saved. The current code accepts it, see 5.2.5	N / Medium	A+
TC-036	BR-02	Deadlines only for enrolled students	SV2 not enrolled in CS101; CS101 has items	1) SV2 opens Assignments 2) SV2 sends POST /api/deadlines/:did/done for a CS101 item	CS101 items are not listed; the request is rejected 403 'Bạn chưa đăng ký môn học này'	S / High	A (test_flow)

F05 - Lecturer Q&A

TC ID	Req / AC	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-037	FR-11 / AC-13	Enrolled student posts a question	SV1 enrolled in CS101	Open Q&A, select CS101, enter title and text, submit	Question appears at the top with author, time and the badge 'Chưa trả lời'	P / High	A (test_flow)
TC-038	FR-11, BR-02 / AC-13	Non-enrolled student cannot post	SV2 not enrolled in CS101	Send POST /api/courses/:cid/questions as SV2 (the UI only offers own courses)	403 'Bạn chưa đăng ký môn học này'; no question created	S / High	A (test_flow)
TC-039	FR-11	Invalid question input	SV1 enrolled in CS101	Submit (a) empty title (b) title of 151 characters (c) text of 2001 characters	Each is rejected with a message (e.g. 'title tối đa 150 ký tự'); nothing saved	N / Medium	A+
TC-040	FR-15 / AC-14	Lecturer sees all questions of the course	GV1 in charge of CS101; 1 answered and 2 unanswered questions	GV1 opens Q&A	All CS101 questions listed; unanswered ones carry the badge and come first; the counter shows 2 unanswered	P / High	M

TC-041	FR-15 / AC-14, BR-09	Lecturer reply marks the question answered	An unanswered question from SV1	1) GV1 types a reply and clicks Send 2) SV1 opens Q&A	The reply is shown with the 'Giảng viên' badge; the question no longer shows 'Chưa trả lời'; SV1 sees the reply	P / High	A (test_flow) + M
TC-042	FR-15, BR-03	Lecturer not in charge	GV2 is not in charge of CS101	1) GV2 opens Q&A 2) GV2 sends a reply via API	CS101 questions are not listed; the reply is rejected 403 'Bạn không thuộc môn học này'	S / High	A (second-lecturer test)
TC-043	BR-09	Student cannot mark own question answered	SV1 has an unanswered question in CS101	SV1 writes a reply under his own question	Per BR-09 the status stays Unanswered. The current code treats any reply as an answer, see 5.2.5	N / High	A+
TC-044	BR-02	Non-enrolled student cannot read Q&A	SV2 not enrolled in CS101	Send GET /api/questions?course_id=CS101 as SV2	403 'Bạn không thuộc môn học này'	S / Medium	A+

Non-functional requirements

TC ID	Req	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-045	NFR-03	Forged or expired token	None	POST /api/login with (a) a token whose payload was changed (b) an expired token (c) a token for another Firebase project	All return 401; no session is created	S / Critical	A (token verification tests)
TC-046	NFR-04	Role enforcement	SV1 and GV1 signed in	As SV1 call every lecturer-only endpoint (create course, add/delete session, add/delete deadline, take over, step down). As GV1 call every student-only endpoint (enroll, cancel, mark done, ask question)	Every call returns 403	S / High	A+
TC-047	NFR-05	Secrets not in the repository	Project repository	1) Open .gitignore and check .env and serviceAccount*.json are listed 2) Run git ls-files and look for .env, service account JSON and *.db files 3) Search the history for 'private_key'	No secret or database file is tracked or found in the history; SECRET_KEY is set in the deployment	S / High	M

TC-048	NFR-07	Vietnamese interface and clear errors	Build under test	Trigger the errors of TC-002, 004, 006, 011, 019, 025 and 038 and read every screen	All text is Vietnamese; each error is one sentence stating the cause; conflict messages name the clashing course and time	U / Medium	M
TC-049	NFR-09	Browser and screen compatibility	Chrome, Edge, Firefox (latest two versions)	Run login, enroll, Timetable, Assignments and Q&A at 360 px, 768 px and 1920 px (DevTools responsive mode)	No overlapping or cut-off elements, no sideways page scrolling, all buttons usable	U / Medium	M
TC-050	NFR-06	First-time student task	5 testers who never used EduMate, a timer	Task: register, enroll in one course, find its sessions in Timetable, without help	At least 4 of 5 finish within 3 minutes	U / Low	M
TC-051	NFR-01, NFR-02	Response and load time	About 50 courses created (repeat the sample data or use a small script)	1) DevTools Network at 10 Mbps, load Courses, Timetable and Assignments 5 times each 2) Optional: 50 parallel requests to GET /api/courses and /api/timetable with k6 or Locust	Each tab loads in 2 s or less (worst of 5 runs); API P95 is 1 s or less	Perf / Low	M

TC-052	NFR-08	Failed operation leaves data unchanged	Test build	Automated: replace the store's create method with one that raises an error, then call enroll. Manual alternative: disconnect the network during Enroll	An error message is shown and no enrollment record exists afterwards	N / Medium	A+
TC-053	NFR-10	Unit test coverage	Repository checked out	Run coverage run -m unittest, then coverage report	All tests pass; the algorithm code (overlap, timetable, deadline classification) is covered at 80% or more (current app.py baseline is about 90%)	P / Medium	A

System (end-to-end) tests

TC ID	Req	Scenario	Preconditions and data	Steps	Expected result	Type / Pri	Method
TC-054	FR-01 to FR-11	Student happy path	Courses CS101, PH101, MA101 exist with deadlines	1) Register a new student 2) Enroll in CS101 3) Enroll in PH101 (back-to-back) 4) Try MA101 (conflict) 5) Open Timetable 6) Open Assignments and mark one item done	Each step behaves as in TC-001, 010, 020, 019, 022, 029, 032, 037 and 007	P / High	M

				7) Ask a question in CS101 8) Log out			
TC-055	FR-01, FR-11 to FR-15	Lecturer happy path	A student enrolled in the new course	1) Register a lecturer 2) Create a course 3) Add a session 4) Add an assignment and an exam 5) The student asks a question 6) The lecturer replies 7) The student reads the reply	Each step behaves as in TC-003, 013, 024, 033, 037 and 041; the student sees the session, the items and the reply	P / High	M

5.2.5 Known gaps between the SRS and the current build

These cases are expected to fail until the team decides whether to change the code or the SRS wording. Log them as defects (S2) or as SRS change requests.

TC	Expected (SRS)	Current behaviour	Action
TC-031	An item due in 3 days is 'Due soon' (AC-11)	The code uses a 72-hour window, so an item due in 3 days at 23:59 can be 'Upcoming'	Choose calendar days or hours, then align code, FR-09 and AC-11
TC-035	A past due date is rejected (BR-06)	Only the date format is checked, so a past date is saved	Add the check, or limit BR-06 to what the system really does
TC-043	Only a lecturer's reply makes a question Answered (BR-09)	Any reply, including a student's, removes the 'Chưa trả lời' badge	Compute the status from the replier's role, or reword BR-09

TC-013	AC-07 says a course is created with its sessions	The course is created with code and name; sessions are added afterwards (TC-024)	Reword FR-12 and AC-07

5.2.6 Proposed acceptance criteria for FR-10 and FR-16

FR-10 and FR-16 have no acceptance criteria in PA2. Proposed wording so that TC-032 and TC-016 to TC-018 have a source:

- **AC-15 (FR-10)** Given an enrolled student sees an unfinished assignment, when they mark it as done, then it is shown as done, listed after the unfinished items and still done after reloading; unmarking restores it.
- **AC-16 (FR-16)** Given a lecturer is not in charge of an existing course, when they take it over, then they become one of the lecturers in charge and can see its questions, while a take-over that overlaps their teaching schedule is rejected with a conflict message; a lecturer who did not create the course can step down, and the creator cannot.

5.2.7 Automation additions

- TC-008 and TC-046: one loop over the endpoint list, expecting 401 without a session and 403 for the wrong role.
- TC-014, TC-026, TC-039: send the invalid input and check the 409 or 400 status.
- TC-021, TC-031, TC-035, TC-043: one request each; TC-031, TC-035 and TC-043 will fail until the decisions in 5.2.5 are made.
- TC-052: pass a store object whose create method raises an exception, then check the response and that no record exists.